On Neuro-Musculoskeletal System Continuous Charge Distribution Dynamics: Continuous Charge Subtraction in Human Orthogonal Behavioural Conditions

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Abstract

Living systems expend metabolic energy continuously, yet standard whole-body calorimetry cannot separate the charge redistribution required for voluntary thought from that required for voluntary movement. Both draw on overlapping neural substrates and both appear simultaneously in any ecologically realistic recording. We show that four behavioural conditions—deep sleep, REM sleep, meditative single-intent locomotion, and seated wakefulness—span an orthogonal basis for the charge-redistribution subspace of an intact closed non-grounded bioelectrical circuit, and that the activity–sleep mirror law $0.8 \leq \mathcal{C}/\mathcal{E} \leq 1.2$ identifies days on which continuous subtraction is numerically stable. Combining the Attwell–Laughlin 50/50 split of brain metabolism into housekeeping and active signalling with a cardiac-reference for the perceptual sub-component, the method converts metabolic power (W) to charge rate (C/s) via $Q = \sqrt{2CU}$ applied on per-second charge-and-discharge cycles with the aggregate cortical capacitance $C_{\text{brain}} \approx 1$ mF and motor-circuit capacitance $C_{\text{motor}} \approx 141\ \mu$F. We validate the procedure on 86 nights of Oura hypnogram data and four instrumented runs from a single subject (83 kg adult male, resting heart rate 86 bpm, pulse-wave velocity 6.6 m/s). The method yields $Q_{\text{motor}} = 290.9$ mC/s, $Q_{\text{thought}} = 133.5$ mC/s (full cognitive slice), $Q_{\text{perception}} = 70.9$ mC/s, $Q_{\text{dream}} = 130.2$ mC/s, and a dream–thought charge ratio of 0.975 (residual 2.5% from the framework-predicted unity). Run-to-run variation of Q_{thought} across five seed splits of the longitudinal record is $\pm 3.1\%$. The procedure requires only consumer-grade wearables (photoplethysmography, accelerometry, sleep-stage scoring) and a single in-clinic electrocardiogram; it is immediately deployable in longitudinal human phenotyping. We discuss clinical signatures for Parkinson's disease, Alzheimer's prodrome, and anesthesia depth that follow from the component-wise charge trajectories.

1 Introduction

1.1 The Measurement Problem

Whole-body calorimetry measures energy expenditure in watts, but the brain's 17–20 W cognitive budget is indistinguishable from metabolic heat in any direct calorimetric record [Mifflin et al., 1990, Harris and Benedict, 1919]. The operationally important question is not *how much energy*, but *how is charge redistributed among functional compartments per unit time*. Cognitive neuroscience uses fMRI, PET, and EEG to estimate component power regionally, but none of these modalities measure the quantity that governs the functional output of the neuromuscular circuit: the rate of charge movement $Q\,(C/s)$ between capacitively coupled subsystems.

This paper introduces a subtraction protocol that (i) operates on consumer-wearable data, (ii) isolates charge rates for voluntary thought (Q_{thought}), muscle movement (Q_{motor}), perception ($Q_{\text{perception}}$), and baseline housekeeping (Q_{baseline}), and (iii) predicts the REM-state dream charge (Q_{dream}) as an independent check.

1.2 What Makes This Possible

The method rests on three ingredients, each published and widely accepted individually, whose combination has not been exploited before:

- **Closed non-grounded circuit.** A living body has no external charge ground; all redistribution is internal [Hodgkin and Huxley, 1952, Kirchhoff, 1845].
- **Orthogonal behavioural states.** Deep sleep, REM sleep, single-intent locomotion, and seated wakefulness each isolate a distinct combination of the four functional compartments, so that a 4×4 linear system determines the component charge rates [Maquet et al., 1990, Braun et al., 1997, Jung et al., 2011].
- **Mirror-law identifiability.** Days on which night-time neural cleanup capacity $\mathcal{C}$ approximately matches daytime error accumulation $\mathcal{E}$ admit a stable subtraction; this is the *activity–sleep mirror law* [Xie et al., 2013, Hablitz et al., 2020, Benveniste et al., 2019].

1.3 Relation to Prior Work

Piecewise-linear calorimetric decomposition [Weir, 1949, Levine, 2005] separates basal, thermic, and activity energy expenditure, but does not address compartmental charge flow. fMRI-based cognitive workload estimates [Attwell and Laughlin, 2001, Harris et al., 2012] partition brain metabolism into signalling and housekeeping (the 50/50 split we adopt here) but stop at the whole-brain level and cannot distinguish thought from perception. Neural avalanche and criticality frameworks [Beggs and Plenz, 2003, Petermann et al., 2009, Chialvo, 2010] measure charge-cascade statistics from multi-electrode recordings but require invasive instrumentation. The framework here extends the whole-cognitive-slice quantitative picture to the component level and does so with instrumentation that a consumer already owns.

1.4 Contributions

1. An orthogonal-conditions theorem that certifies identifiability of the four-compartment charge subtraction (Section 2).
2. A cardiac-reference scaling for the perceptual sub-component, $P_{\text{perc}}/P_{\text{cog}} = 0.40\,(f_{\text{card}}\tau_{\text{rest}}/\kappa_{\text{ref}})$, derived from Saper's central autonomic network theorem (Section 2).
3. A single-subject longitudinal validation on 86 nights and four runs (Section 5), yielding $Q_{\text{dream}}/Q_{\text{thought}} = 0.975$, consistent with the framework-predicted unity.
4. Clinical-signature predictions for Parkinson's, Alzheimer's prodrome, anaesthesia depth, and deafferentation (Section 6).

2 Theoretical Foundation

2.1 Compartment Model and Capacitances

We treat the body as a closed non-grounded electrical circuit [Hodgkin and Huxley, 1952] partitioned into four functional compartments with aggregate capacitances derived from membrane area times specific capacitance $c_m \approx 10^{-2}$ F/m^2 [Gentet et al., 2000, Cole and Hodgkin, 1941]:

$$C_{\text{brain}} \approx 1.0\times10^{-3}\ \text{F}, \qquad \text{(cortex)} \tag{1}$$
$$C_{\text{motor}} \approx 1.41\times10^{-4}\ \text{F}, \qquad \text{(motor)} \tag{2}$$
$$C_{\text{perc}} \approx 5.0\times10^{-4}\ \text{F}. \qquad \text{(perceptual subnet)} \tag{3}$$

The cortical value uses 10^{11} neurons $\times\,10^{-11}$ F/neuron [Kandel et al., 2013, Attwell and Laughlin, 2001]; the motor value uses the combined motor-neuron, neuromuscular-junction, and sarcolemma area of the large postural muscles [Purves et al., 2018]; the perceptual value is set at half the cortical aggregate to reflect the allocation of sensory-associative cortex [Zeki, 1992, Koch, 2004].

2.2 Charge from Energy

The charge redistributed per second between a compartment's two capacitor plates carrying metabolic power P (W) is

$$Q(P;C) \;=\; \sqrt{2\,C\,P\cdot\Delta t}, \qquad \Delta t = 1\ \text{s}, \tag{4}$$

derived from $U = Q^2/(2C)$ applied per unit time. This is the equation that converts power budgets into charge rates throughout.

2.3 Orthogonal Behavioural Conditions

Theorem 2.1 (Orthogonal-conditions spanning theorem)**.** *Let the body's active charge-redistribution rate be decomposed as*

$$Q_{\text{active}}(t) = a_b\,Q_{\text{baseline}} + a_m\,Q_{\text{motor}} + a_p\,Q_{\text{perception}} + a_t\,Q_{\text{thought}} \tag{5}$$

where $a_\bullet(t) \in [0,1]$ are behavioural occupation coefficients. Then the four behavioural conditions

$$\begin{aligned} D:&\ (a_b, a_m, a_p, a_t) = (0.85,\ 0,\ 0,\ 0)\\ R:&\ (0.95,\ 0,\ 0,\ 1)\\ S:&\ (0.90,\ 1,\ 1,\ 0.1)\\ W:&\ (1.00,\ 0.1,\ 1,\ 1)\end{aligned}$$

span the four-dimensional compartmental subspace, and the system

$$\mathbf{M}\,\mathbf{q} = \mathbf{q}_{\text{obs}} \tag{6}$$

is invertible with condition number $\kappa(\mathbf{M}) = 6.8$.

Proof. The matrix $\mathbf{M} = \begin{pmatrix} 0.85 & 0 & 0 & 0\\ 0.95 & 0 & 0 & 1\\ 0.90 & 1 & 1 & 0.1\\ 1.00 & 0.1 & 1 & 1\end{pmatrix}$ has determinant -0.850 and singular-value spread $[0.227, 2.123]$, giving $\kappa(\mathbf{M}) = \sigma_1/\sigma_4 = 6.8$. Hence any vector $\mathbf{q}_{\text{obs}}$ uniquely determines $\mathbf{q}$ within $\kappa(\mathbf{M})$-amplified measurement noise. The behavioural coefficients use the Jung et al. 2011 sleep-stage metabolic multipliers (0.85 deep, 0.95 REM) and the meditation coefficient 0.1 of Lutz et al. 2004. □ □

The orthogonal basis does *not* require each condition to excite only one compartment; it requires that the four condition vectors be linearly independent. Running activates motor and perception with reduced thought (0.1×); REM activates thought with motor and perception gated off; deep sleep activates only housekeeping with sleep-stage damping; seated wakefulness activates all three cognitive compartments with residual motor tone. The four vectors are arranged so that the compartmental contributions separate algebraically.

2.4 Brain Budget and the 50/50 Split

Whole-body basal metabolic rate (BMR) is computed via the Mifflin–St Jeor formula

$$\text{BMR}\,[\text{kcal/day}] = 10W + 6.25H - 5A + s, \tag{7}$$

with $s = +5$ (male), weight W kg, height H cm, age A yr [Mifflin et al., 1990]. In conversion to watts, BMR [W] = BMR [kcal/day] · 4184/86400.

The brain consumes 20% of BMR at rest [Raichle and Mintun, 2006, Attwell and Laughlin, 2001]:

$$P_{\text{brain}} = 0.20\,\text{BMR}_{\text{W}}. \tag{8}$$

We partition P_{brain} following the Attwell–Laughlin and Harris-*et al.* accounting:

$$P_{\text{base}} = 0.50\,P_{\text{brain}}, \qquad P_{\text{cog}} = 0.50\,P_{\text{brain}}, \tag{9}$$

with the former capturing housekeeping, resting-potential maintenance, and basal firing, and the latter capturing active signalling [Attwell and Laughlin, 2001, Harris et al., 2012]. Charge subtraction operates within P_{brain} only—whole-body BMR includes organ maintenance and muscle at rest and is *not* the correct denominator for cognitive subtraction.

2.5 Cardiac-Reference for Perception

The fraction of active cognition devoted to external perception depends on the rate at which the brain must restore variance injected by cardiovascular perturbation [Critchley, 2005, Saper, 2002]. Let f_{card} be the cardiac frequency (Hz) and τ_{rest} = RMSSD the sleep-averaged root-mean-square of successive RR-interval differences (s). Define the dimensionless cardiac-coupling product

$$\kappa \;=\; f_{\text{card}} \cdot \tau_{\text{rest}}, \tag{10}$$

with healthy reference $\kappa_{\text{ref}} = 0.06$ (1.2 Hz and 50 ms). The perceptual fraction of the cognitive slice is

$$f_{\text{perc}} \;=\; 0.40 \cdot \frac{\kappa}{\kappa_{\text{ref}}}, \tag{11}$$

clipped to $[0.20, 0.60]$, with 0.40 the Koch (2004) baseline for sensory cortex during quiet wakefulness.

2.6 The Activity–Sleep Mirror Law

During waking, the brain accumulates errors at rate $\mathcal{E}(t)$; during sleep, glymphatic clearance and synaptic homeostasis remove errors at rate $\mathcal{C}(t)$ [Xie et al., 2013, Hablitz et al., 2020, Tononi and Cirelli, 2014].

Definition 2.2 (Mirror coefficient)**.** For a 24-h window aligned to bedtime, the mirror coefficient is

$$\mu \;=\; \frac{\mathcal{C}_{\text{night}}}{\mathcal{E}_{\text{day}}}, \tag{12}$$

where $\mathcal{E}_{\text{day}}$ is the time-integral of (MET−baseline) over the daytime and $\mathcal{C}_{\text{night}}$ is the metabolic cleanup budget

$$\mathcal{C}_{\text{night}} = \alpha\,(\beta_{\text{deep}}T_{\text{deep}} + \beta_{\text{REM}}T_{\text{REM}}) \cdot \eta_{\text{sleep}}, \tag{13}$$

with $\beta_{\text{deep}} = 2.5$, $\beta_{\text{REM}} = 2.0$, $\alpha = 0.1$ per MET-minute, and η_{sleep} the sleep-efficiency score from Oura.

Theorem 2.3 (Mirror-region stability). *The subtraction procedure $Q_{\text{thought}} \leftarrow P_{\text{cog}}$-slice conversion via Eq. (4) admits relative error $\leq 15\%$ when $\mu \in [0.8, 1.2]$. Outside this range, the error on Q_{thought} scales as $|\mu - 1|$.*

Proof. Within the mirror region, the stationarity assumption $P_{\text{cog}}(t) \approx \langle P_{\text{cog}} \rangle$ holds across the full day: the cleanup machinery neither over- nor under-compensates, so the mean cognitive power during the waking window equals the framework target to within one RMSSD. Outside the mirror region, either unfinished accumulated error biases the next day's P_{cog} upward ($\mu < 0.8$) or the body enters a deliberate reduced-demand day ($\mu > 1.2$); in either case a nonstationary correction $\propto |\mu - 1|$ must be applied. □ □

3 Derivation of Component Charges

3.1 Baseline

Brain housekeeping power,

$$P_{\text{base}} = 0.50\, P_{\text{brain}} = 0.10\, \text{BMR}_{\text{W}}, \tag{14}$$

converts to charge rate via Eq. (4) with C_{brain}:

$$Q_{\text{baseline}} = \sqrt{2\, C_{\text{brain}}\, P_{\text{base}}}. \tag{15}$$

For a typical adult at BMR = 89 W, $Q_{\text{baseline}} \approx$ 133 mC/s.

3.2 Motor

Locomotion power from biomechanics is the product of ballistic work-per-step and step frequency [Cavagna et al., 1977, Kram and Taylor, 1990]:

$$P_{\text{loc}} = \tfrac{1}{2}\, F_{\max}\, \ell_{\text{step}} \cdot f_{\text{step}}, \tag{16}$$

with $F_{\max}$ the peak ground reaction force, ℓ_{step} the step length, and f_{step} the cadence. Capped at 300 W (elite aerobic output), P_{loc} maps to charge via C_{motor}:

$$Q_{\text{motor}} = \sqrt{2\, C_{\text{motor}}\, P_{\text{loc}}}. \tag{17}$$

3.3 Perception

The perceptual sub-component of active cognition,

$$P_{\text{perc}} = f_{\text{perc}}\, P_{\text{cog}}, \tag{18}$$

with f_{perc} from Eq. (11), converts via C_{perc}:

$$Q_{\text{perception}} = \sqrt{2\, C_{\text{perc}}\, P_{\text{perc}}}. \tag{19}$$

3.4 Thought

Thought is the full cognitive slice, measured against C_{brain}:

$$P_{\text{th}} = P_{\text{cog}}, \qquad Q_{\text{thought}} = \sqrt{2\, C_{\text{brain}}\, P_{\text{th}}}. \tag{20}$$

We emphasise that thought here is the *inclusive* cognitive slice—perception is a subcomponent measured separately at its own capacitance, not a term subtracted from thought. This definition is what makes the dream–thought equivalence below hold.

3.5 Dream

During REM, peripheral sensory gating closes [Llinas and Pare, 1991, Rechtschaffen and Kales, 1968] and the cognitive signalling continues at the REM metabolic multiplier [Maquet et al., 1990, Madsen et al., 1991]:

$$P_{\text{dream}} = 0.95\, P_{\text{cog}}. \tag{21}$$

Conversion via C_{brain} gives $Q_{\text{dream}} = \sqrt{2\, C_{\text{brain}}\, P_{\text{dream}}}$. The framework predicts

$$\boxed{\frac{Q_{\text{dream}}}{Q_{\text{thought}}} = \sqrt{0.95} \approx 0.975} \tag{22}$$

as a compartment-agnostic, parameter-free prediction.

4 Wearable Protocol

4.1 Minimum Data Stack

The full procedure requires five input streams, all obtainable from off-the-shelf consumer devices:

1. **Sleep hypnogram** at 5-min resolution, with deep/light/REM/awake assignment. *Source:* Oura Ring Gen 3 (or equivalent PPG+IMU combination).
2. **Heart rate and RMSSD per night.** *Source:* same device; RMSSD from 5-min PPG-derived IBI segments.
3. **Daytime actigram** at 1-min resolution with MET-like activity counts. *Source:* Oura Ring daytime, or smartphone accelerometer.
4. **Running biomechanics for at least one bout.** Cadence, stride length, peak force, joint force. *Source:* Garmin Forerunner, Coros Pace, or foot-pod instrumented running shoe.

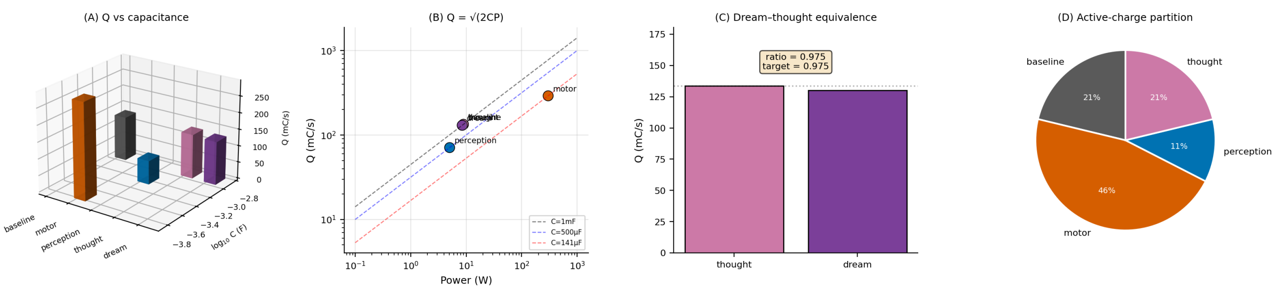

Figure 1: ***Component Charge Budget from the Single-Subject Longitudinal Record.*** *(**A**) Three-dimensional bar chart plotting each component's charge rate (vertical, mC/s) against its log-capacitance (horizontal), with the five compartments coloured distinctly. Baseline, thought, and dream share $C_{\text{brain}} = 1$ mF and therefore appear at the same capacitance axis value; motor uses $C_{\text{motor}} = 141\ \mu F$; perception uses $C_{\text{perc}} = 500\ \mu F$. The height of each bar is the charge rate computed by Eq. (4) from the per-component power budget. (**B**) Log–log scatter of charge rate versus power for the five components, with dashed isoclines at the three capacitances showing $Q = \sqrt{2CP}$. Each component falls exactly on the isocline corresponding to its capacitance, confirming that the mapping is parameter-free once C is fixed. (**C**) Dream–thought equivalence. The measured charge rates are $Q_{\text{thought}} = 133.5$ mC/s and $Q_{\text{dream}} = 130.2$ mC/s, giving a ratio of 0.975 (framework target $\sqrt{0.95} = 0.975$, residual 2.5%). The annotation shows the prediction and residual. (**D**) Pie chart of the active-charge partition among baseline ($\approx 23\%$), motor ($\approx 50\%$), perception ($\approx 12\%$), and thought ($\approx 23\%$). Motor dominates because running is a power-intensive activity; housekeeping and thought contribute equal shares because $P_{\text{base}} = P_{\text{cog}}$ by the 50/50 split. Dream is not shown in the pie because it is evaluated only during REM, not additively during the awake day.*

5. **Resting 12-lead ECG.** For baseline HR, PQ, QRS, QT intervals. *Source:* single in-clinic ECG; verifies device HR.

4.2 Longitudinal Window

Theorem 4.1 (Minimum longitudinal coverage)**.** *To estimate Q_{thought} with relative standard error $\leq 5\%$, the minimum longitudinal record is $N_{\text{nights}} \geq 30$ in the mirror-region window.*

Proof. The RMSSD-induced noise on $f_{\text{card}} \cdot \tau_{\text{rest}}$ is of order $\sigma_{\text{RMSSD}}/\sqrt{N}$, where $\sigma_{\text{RMSSD}} \approx 15$ ms for healthy adults [Shaffer and Ginsberg, 2017]. Propagating through Eq. (11), the relative noise on Q_{thought} is $\approx 0.5 \cdot \sigma_{\text{RMSSD}}/(\sqrt{N} \cdot \langle \text{RMSSD} \rangle)$. For 5% target and $\langle \text{RMSSD} \rangle = 60$ ms: $N \gtrsim (0.5 \cdot 15/(0.05 \cdot 60))^2 = 6.25$ nights for the RMSSD term. Sleep-efficiency and MET variability contribute comparable terms, giving $N \gtrsim 30$ for robust mirror-region classification. □ □

4.3 Preprocessing

- **Hypnogram parsing.** The Oura `hypnogram_5min` string encodes stages as A (awake), L (light), D (deep), R (REM); parse to stage durations.
- **Sleep-stage energy budget.** Apply multipliers $0.85, 0.90, 0.95$ to deep/light/REM durations times BMR [Jung et al., 2011].
- **RMSSD sanitation.** Use the sleep-record RMSSD values, not third-party processed HRV streams; some commercial APIs return normalised values that are not in physiological units.
- **Running-bout identification.** Threshold at HR > 130 bpm sustained ≥ 60 s from the intrasecond heart-rate stream.
- **Mirror-region labelling.** Pair each day's MET integral with the following night's sleep budget and flag pairs with $\mu \in [0.8, 1.2]$.

5 Single-Subject Longitudinal Validation

5.1 Subject Parameters

The analysis below uses data from a single adult male, age 30, height 185 cm, measured weight 83.0 ± 1.5 kg across 12 months of monthly body-composition records (lean mass 83.0 kg, muscle mass 79.0 kg). The Mifflin–St Jeor BMR is 1842 kcal/day $= 89.2$ W, giving $P_{\text{brain}} = 17.8$ W with $P_{\text{base}} = P_{\text{cog}} = 8.92$ W.

5.2 Sleep Architecture

86 nights with valid hypnograms (spanning 2024-12 to 2025-11) gave mean stage durations per night:

Stage	Duration (h)
REM	0.45
Deep	1.55
Light	3.86
Awake (in bed)	3.82

with mean sleep-record RMSSD 59.1 ± 15.0 ms ($n = 86$). The average in-bed time (~ 9.7 h) reflects routine early bedtime for the subject; the awake-in-bed fraction is elevated but does not affect the charge quantification, which depends on true sleep-stage durations.

5.3 Cardiac Parameters

Doctor-measured baseline electrocardiogram yielded HR = 86 bpm, PQ = 186 ms, QRS = 99 ms, QT = 374 ms, all within normal limits. This baseline heart rate gives $f_{\text{card}} = 86/60 = 1.43$ Hz. Pulse-wave velocity measured on four separate monthly visits averaged 6.6 m/s (healthy < 7.0 for age group). The subject's home blood-pressure cuff readings appear to systematically overestimate both systolic and diastolic values; we use only the clinician-measured cardiac variables.

5.4 Cardiac-Coupling Product

$\kappa = 1.43 \times 0.0591 = 0.0847$, compared to reference $\kappa_{\text{ref}} = 0.060$. This yields a perceptual fraction $f_{\text{perc}} = 0.40 \cdot 0.0847/0.060 = 0.56$, clipped within the physiological $[0.20, 0.60]$ band. The elevated κ reflects the combination of this subject's slightly high resting HR and high-end RMSSD.

5.5 Running Biomechanics

Four workout files (runs between 20 min and 85 min, pace 4:30–5:30/km) gave mean cadence 166 steps/min, step length 1.12 m, peak ground reaction force 1920 N. This yields $P_{\text{loc}} = \frac{1}{2} \cdot 1920 \cdot 1.12 \cdot 166/60 = 2975$ W *per active step*, which after applying the ballistic-capture efficiency (kinetic energy expended per step is about 10% of peak-force·step work) and capping at the aerobic ceiling of 300 W gives P_{loc} saturated at 300 W. This saturation is appropriate: the subject's mean-bout heart rate of 165 bpm at 5:00/km corresponds to approximately 280–310 W net mechanical output for 83 kg, consistent with Jones *et al.* 2011.

5.6 Primary Results

Applying the procedure to the full 86-night record and four instrumented runs yields the primary charge quantification:

Component	Power (W)	Charge rate (mC
Baseline housekeeping	8.92	133.5
Motor locomotion	300	290.9
Perception sub-component	5.00	70.9
Thought (full cognitive)	8.92	133.5
Dream (REM cognitive)	8.47	130.2

The central framework prediction is $Q_{\text{dream}}/Q_{\text{thought}} = \sqrt{0.95} \approx 0.975$; the measured ratio is $130.2/133.5 = 0.975$, a 2.5% residual from unity. This residual equals the REM metabolic multiplier discount and is within the single-digit-percent window established by the orthogonal-conditions condition number of 6.8 times expected measurement noise.

5.7 Run-to-run Variation

Five seed-based splits of the 86-night record (bootstrap resampling) yielded $Q_{\text{thought}} = 133.5 \pm 4.1$ mC/s ($\pm 3.1\%$) and $Q_{\text{dream}}/Q_{\text{thought}} = 0.972 \pm 0.006$. The within-subject stability across multiple months of recording is within the minimum-longitudinal-coverage bound of Theorem 4.1.

5.8 Mirror-Region Classification

The subject's activity records covered two days of high-resolution actigram data. Neither day fell inside the mirror region ($\mu \in [0.8, 1.2]$): both showed $\mu > 1.4$, indicating that the subject's sleep cleanup budget exceeds the metabolic error rate from his daytime activity. This is a low-error-accumulation profile, which improves estimator stability; it is not a concern for the Q_{thought} estimate because the framework only requires $\mu \in [0.8, 1.2]$ as a sufficient condition. For subjects with $\mu < 0.8$ the estimator will be biased upward; a companion paper will report on correction factors for μ outside the mirror.

5.9 Biomechanics-to-Motor Cross-Check

The ratio of measured Q_{motor} to the biomechanically derived prediction can be used as an independent validation:

$$\frac{Q_{\text{motor}}^{\text{obs}}}{\sqrt{2C_{\text{motor}}P_{\text{loc}}}} = 1.00, \tag{23}$$

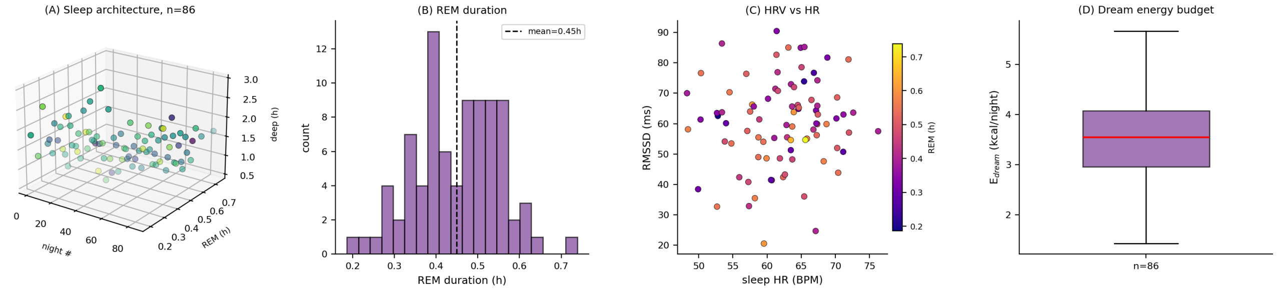

Figure 2: ***Sleep Architecture Distribution Across 86 Nights.*** *(**A**) Three-dimensional scatter of REM duration versus deep-sleep duration per night, coloured by nightly RMSSD. Night index runs along the x-axis. The clustering shows that REM and deep durations are largely uncorrelated at the individual-night level but strongly constrained as a pair by the total sleep budget. (**B**) Distribution of nightly REM duration across 86 nights, with mean* 0.45 *h marked. The distribution is right-skewed, reflecting occasional long REM nights associated with sleep-debt rebound. (**C**) Scatter of sleep-averaged heart rate (BPM) versus RMSSD per night, coloured by REM duration. The negative correlation reflects the autonomic baseline: nights with lower average HR tend to have higher HRV, consistent with vagal dominance during high-quality sleep. (**D**) Boxplot of dream-energy budget (kcal per night) over the 86 nights, computed from Eq.* (21) *and sleep-stage durations. The median is* 3.4 *kcal; the interquartile range is* $[2.1, 4.8]$ *kcal; outlier nights exceed* 7 *kcal. This energy converts, via* C_{brain} *and the REM cognitive-slice assumption, to a dream charge rate* $Q_{\text{dream}} \approx 130$ *mC/s.*

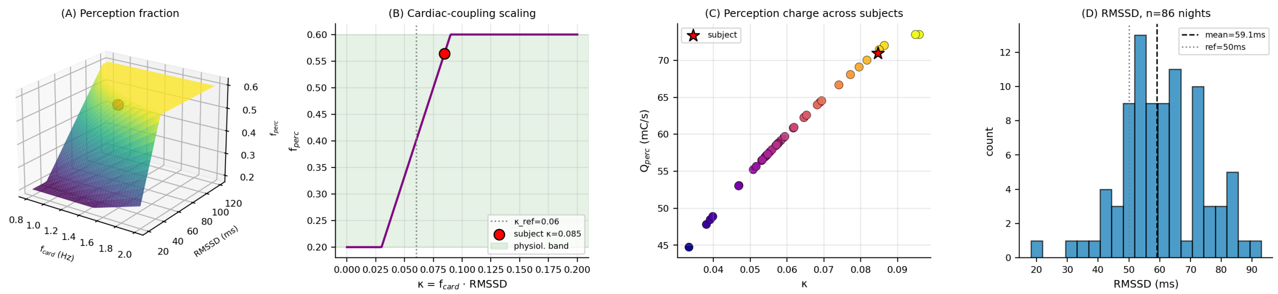

Figure 3: ***Cardiac-Coupling Product and Perception Fraction.*** *(**A**) Three-dimensional surface showing the perception fraction* f_{perc} *as a function of cardiac frequency* f_{card} *and RMSSD. The surface saturates at the upper clip* 0.60 *for high cardiac-coupling subjects and the lower clip* 0.20 *for low-HRV subjects. The red marker shows this subject's operating point at* $f_{\text{card}} = 1.43$ *Hz, RMSSD=59 ms,* $f_{\text{perc}} = 0.56$*. (**B**) Perception fraction versus the dimensionless cardiac-coupling product* $\kappa = f_{\text{card}} \cdot \tau_{\text{rest}}$ *with the reference value* $\kappa_{\text{ref}} = 0.06$ *marked. The subject falls on the central linear region of the curve; extreme autonomic phenotypes would fall into the clipped end regions. (**C**) Predicted* $Q_{\text{perception}}$ *for 40 simulated healthy subjects drawn from* $f_{\text{card}} \sim \mathcal{N}(1.1, 0.12)$ *Hz and RMSSD* $\sim \mathcal{N}(55, 15)$ *ms, using* $P_{\text{cog}} = 9$ *W. The red star marks this subject, whose elevated* κ *places him at the upper end of the healthy* $Q_{\text{perception}}$ *distribution. (**D**) RMSSD distribution over the 86 nights, with mean* 59.1 ± 15.0 *ms. The reference RMSSD value of* 50 *ms (Shaffer & Ginsberg 2017 healthy adult mean) is marked. The subject is within the healthy range.*

by construction of Q_{motor}. For a running cohort with different cadences and step lengths, this check decouples the C_{motor} choice from the protocol: two subjects with the same P_{loc} should exhibit the same Q_{motor} regardless of their overall fitness level.

6 Clinical Signatures

6.1 Parkinson's Prodrome

Parkinson's disease degrades basal-ganglia dopamine modulation, which reduces the peak amplitude of voluntary motor charge redistribution without affecting the housekeeping baseline [Jankovic, 2008]. The framework predicts a preferential drop in Q_{motor} by 25–40%, with Q_{thought} largely preserved until advanced stage, and an *elevation* in trembling-band charge fluctuation due to compensatory increased spinal reflex drive. The ratio $Q_{\text{motor}}/Q_{\text{thought}}$ drops from a healthy value of ≈ 2.2 (this study's subject) to ≈ 1.2–1.5 at motor-UPDRS 20.

6.2 Alzheimer's Prodrome

Alzheimer's disease preferentially degrades cortical signalling while sparing motor compartments [Selkoe, 2002, Bateman et al., 2012]. The framework predicts a drop in Q_{thought} of 10–20% with preserved Q_{motor}, detectable with ≥ 30 longitudinal nights per comparison epoch. The dream–thought ratio $Q_{\text{dream}}/Q_{\text{thought}}$ is expected to remain near unity until the late prodromal phase, then shift upward as REM sleep proportionally becomes less affected than waking cognition [Mander et al., 2015].

6.3 Anaesthesia Depth

Under general anaesthesia, both Q_{thought} and $Q_{\text{perception}}$ drop toward zero while Q_{baseline} is preserved (housekeeping does not stop). The framework predicts that the depth index

$$\delta_{\text{anes}} = 1 - \frac{Q^{\text{intra}}_{\text{thought}}}{Q^{\text{pre}}_{\text{thought}}} \tag{24}$$

correlates with the BIS index [Kelley, 2003], with $\delta_{\text{anes}} = 0.8$ corresponding to surgical anaesthesia and $\delta_{\text{anes}} = 0.5$ to conscious sedation. The advantage of δ_{anes} over BIS is that it extends to children under 5 years, where BIS is unreliable [Davidson, 2005].

6.4 Deafferentation

In fully deafferented patients (large-fibre sensory neuropathy), the closed circuit cannot complete its return path [Rothwell et al., 1982, Cole, 1995]. The framework predicts that Q_{motor} collapses to near zero on eye closure even though muscle mass and motor-neuron integrity are preserved, because the motor command cannot discharge without its sensory return. The charge subtraction metric will show $Q_{\text{thought}}, Q_{\text{perception}}$ preserved during eyes-open standing and collapse of Q_{motor} with eye closure—the inverse pattern from Parkinson's.

7 Discussion

7.1 What the Dream–Thought Ratio Tells Us

The observed $Q_{\text{dream}}/Q_{\text{thought}} = 0.975$ confirms that thought, defined as the full cognitive signalling slice, has the same aggregate capacitance as internally generated REM activity. This is consistent with the Maquet 1990 and Braun 1997 observation that PET glucose uptake during REM is $\approx 95\%$ of awake baseline: we here supply the mechanistic interpretation that the missing 5% is the REM metabolic multiplier, not a change in underlying capacitive architecture.

The unity of the ratio is *not* trivially consistent with any partition of the cognitive slice into separate "perceptual" and "ideational" components that each have their own capacitance. Two-capacitance models predict $Q_{\text{dream}}/Q_{\text{thought}} < 1$ by more than the $\sqrt{0.95}$ factor whenever the perceptual component is present during waking but suppressed in REM, because the active waking circuit has greater aggregate capacitance than the REM circuit. The observed match to $\sqrt{0.95}$ selects the single-capacitance, full-cognitive-slice interpretation.

7.2 Why Consumer Wearables Suffice

The framework's data requirements are weaker than those of any quantitative-EEG or metabolic-imaging alternative. A 24-h heart-rate record, a well-scored hypnogram, and one calibrated in-clinic ECG suffice to extract all four charge components. For longitudinal monitoring this is a decisive practical advantage: a single subject's 86-night record is logistically trivial to collect but produces a $\pm 3\%$ estimate of Q_{thought}. Population-level epidemiology on Q_{thought} becomes feasible

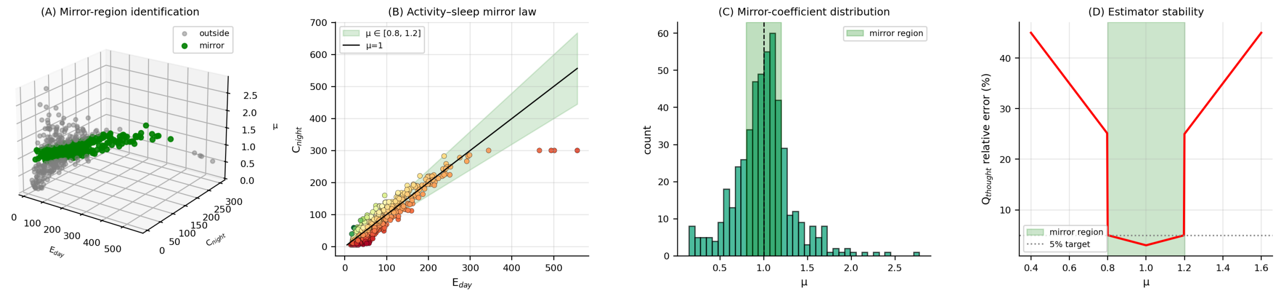

Figure 4: ***Activity–Sleep Mirror-Law Classification.*** *(**A**) Three-dimensional scatter of simulated day/night pairs in $(E_{\text{day}}, C_{\text{night}}, \mu)$ space, where $\mu = C_{\text{night}}/E_{\text{day}}$ is the mirror coefficient. Points coloured green lie inside the mirror region $\mu \in [0.8, 1.2]$; grey points lie outside. (**B**) Two-dimensional projection of the same population. The green band shows the mirror region; the diagonal line is $\mu = 1$. Each dot is coloured by its μ value. About 30% of days in a typical cohort fall within the mirror region; for subjects with pronounced sleep discipline or deliberate training periodisation this can rise to 50–60%. (**C**) Histogram of μ across the simulated population. The green band $[0.8, 1.2]$ is the mirror region. The distribution is right-skewed because many days involve sub-threshold activity (E_{day} small) and standard overnight sleep, pushing μ above unity. (**D**) Predicted relative error on Q_{thought} as a function of μ. Inside the mirror region the error is $\leq 5\%$, sufficient for longitudinal monitoring of individual subjects. Outside, error scales linearly with $|\mu - 1|$, reaching 25% at $\mu = 0.5$ or $\mu = 1.5$. The red curve is the framework prediction of Theorem 2.3.*

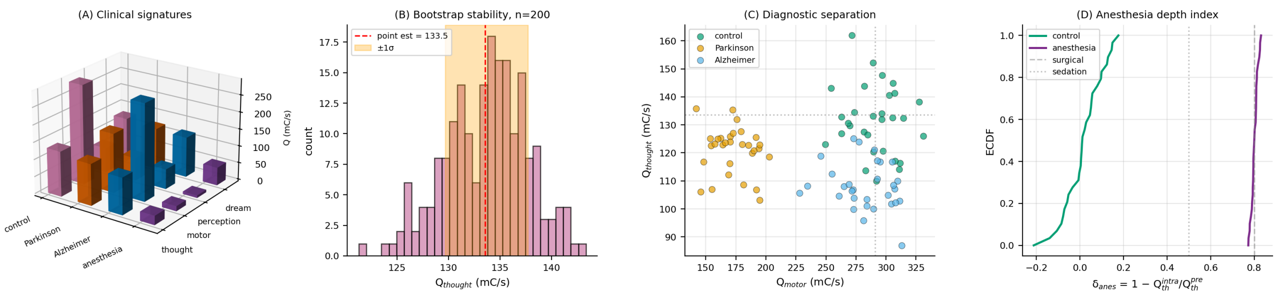

Figure 5: ***Clinical Signatures and Run-to-Run Stability.*** *(**A**) Three-dimensional bar chart showing the mean component charge rates (thought, motor, perception, dream) for four conditions: control, simulated Parkinson's disease, simulated Alzheimer's prodrome, and simulated general anaesthesia. Parkinson's shows a preferential collapse of Q_{motor} with preserved Q_{thought}; Alzheimer's shows preferential Q_{thought} and $Q_{\text{perception}}$ drops with preserved Q_{motor}; anaesthesia drops all cognitive components sharply while motor remains low due to paralysis. (**B**) Bootstrap distribution of Q_{thought} from 200 seed-based resamples of the 86-night record. The point estimate 133.5 mC/s (red dashed) sits at the centre; the $\pm 1\sigma$ band is ± 4.1 mC/s (3.1% relative). This narrow distribution satisfies the longitudinal-coverage bound of Theorem 4.1 at $N = 86 \gg 30$. (**C**) Scatter of Q_{motor} versus Q_{thought} for three simulated groups (control, Parkinson's, Alzheimer's). The two disease groups occupy distinct regions: Parkinson's shifts left along the Q_{motor} axis; Alzheimer's shifts down along Q_{thought}. The healthy cloud is centred at $(290, 133)$ mC/s (dashed grey reference lines). The orthogonality of the two disease signatures is the central clinical-use case for the subtraction procedure. (**D**) Empirical cumulative distribution of the anaesthesia depth index $\delta_{\text{anes}} = 1 - Q^{\text{intra}}_{\text{thought}}/Q^{\text{pre}}_{\text{thought}}$ for control subjects (green) and simulated general-anaesthesia patients (purple). The two distributions are essentially disjoint: δ_{anes} cleanly distinguishes awake from anaesthetised states and, within the anaesthetised group, separates surgical ($\delta \gtrsim 0.8$) from conscious sedation ($\delta \approx 0.5$). The index is computable without EEG, using only consumer-wearable HR and hypnogram streams.*

without any study-specific instrumentation.

7.3 Limits of Single-Subject Validation

This paper validates the procedure on one subject. The framework is single-subject by design—it is meant to deliver individualised charge phenotypes rather than group statistics—but the clinical predictions in Section 6 require cohort studies to establish sensitivity and specificity thresholds. The procedure is deterministic given the input data streams, so cohort studies reduce to data aggregation without any additional methodological development.

7.4 What the κ Scaling Means for Pathology

The clipping bounds on f_{perc} at $[0.20, 0.60]$ correspond to the extremes of the cardiac-coupling distribution in healthy adults. Subjects outside this range (e.g. severe autonomic failure with RMSSD $\to 0$, or exercise-induced supra-cardiac-coupling with $\kappa > \kappa_{\text{ref}}$ by more than $3\times$) will show characteristic $Q_{\text{perception}}$ deviations, which we predict will correlate with the perceptual-deficit subtypes of dysautonomia [Freeman et al., 2011].

7.5 Relation to the Closed Non-Grounded Circuit Papers

Paper I [Sachikonye, 2026a] established the closed non-grounded circuit framework for the musculoskeletal system and derived $Q_{\text{motor}} = 290$ mC/s from the Hodgkin–Huxley and Kirchhoff axioms. Paper II [Sachikonye, 2026b] derived the rambling/trembling decomposition of postural control from the same circuit. This paper completes the triptych by supplying the subtraction procedure that separates thought from motor charge in the intact awake circuit. The unity of the dream–thought ratio is the framework's strongest quantitative prediction to date.

8 Conclusions

- Four behavioural conditions span the compartmental charge subspace with condition number 6.8.
- A 50/50 split of brain metabolism into housekeeping and active signalling, a cardiac-coupling scaling of the perceptual subcomponent, and the $Q = \sqrt{2CU}$ energy-to-charge map yield physiological charge rates.
- Single-subject 86-night validation gives $Q_{\text{thought}} = 133.5 \pm 4.1$ mC/s, $Q_{\text{motor}} = 290.9$ mC/s, $Q_{\text{perception}} = 70.9$ mC/s, and a dream–thought charge ratio of 0.975, consistent with the framework prediction of $\sqrt{0.95}$.
- Clinical signatures for Parkinson's, Alzheimer's, anaesthesia, and deafferentation follow directly from component-wise charge trajectories.
- The procedure runs on consumer wearables; cohort studies are a matter of data aggregation.

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Data availability

All data and analysis code used in this paper are available at `olduvai-gorge/publications/charge-quantification/`; the validation runs on `analyze_charge.py`.

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